

Single Port Ram Verification Plan

1. DUT Overview & Interface

The Design Under Test (DUT) is a parameterized **synchronous single-port RAM** named **simple_ram**.

- **Parameters:** **ADDR_WIDTH** (default 4), **DATA_WIDTH** (default 4), **DEPTH** (default 16).
- **Control:** Chip Select (cs), Write Enable (we).
- **Behavior:**
 - Synchronous writes occur on the positive clock edge when both cs and we are high.
 - Synchronous reads occur on the positive clock edge when cs is high and we is low.

2. Features to be Verified

- **Write Operations:** Data is successfully written to the target memory array location when selected.
- **Read Operations:** Data is successfully retrieved from the memory array when selected.
- **Chip Select Logic:** The memory array ignores all read/write attempts when cs is low.
- **Data Integrity:** Data is preserved across addresses without leakage or overwriting adjacent memory spaces.

3. Test Scenarios (Stimulus)

Test Name	Description	Priority
Basic Read/Write	Write a known value to a specific address, disable cs, enable cs, and read it back.	High
Memory Fill & Dump	Sequentially write unique data to all addresses from 0 to DEPTH-1 , then sequentially read them all back.	High
Back-to-Back Writes	Perform consecutive writes to different addresses without dropping cs or we.	Medium
Read-After-Write (RAW)	Write to an address and	High

	immediately read from the same address on the next clock cycle.	
Unselected Operations	Drive toggling we, addr, and din while keeping cs low. Verify memory contents remain completely unchanged.	High
Randomized Stress Test	Fully randomized sequence of cs, we, addr, and din to hit crossing scenarios.	Medium

4. Checking Methodology (Scoreboarding)

To verify the RAM accurately, the UVM scoreboard should implement a "Shadow Memory" predictor model.

- **Predictor Logic:** On a valid write transaction (cs == 1, we == 1), the scoreboard updates its internal shadow memory.
- **Comparison Logic:** On a valid read transaction (cs == 1, we == 0), the scoreboard compares the DUT's dout against the data stored in its shadow memory at the requested addr.

5. Functional Coverage Plan

- **Control Signals Toggle:** Ensure cs and we hit both 0 and 1 states.
- **Address Space Exhaustion:** Verify every valid address (cp_addr_bin) has been targeted.
- **Data Space Exhaustion:** Verify every valid data value (cp_data_bin) has been driven.
- **Cross Coverage:**
 - cs == 1 crossed with we == 1/0 across all addresses.
 - Ignore conditions where cs == 0 for memory space coverage, as the DUT ignores these inputs.